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Abstract

We present the Parametric Trotter Expression (PTE) family of operator primitives — Global PTE, Product PTE, and Local PTE (LPTE) — and show that they form a hierarchy of increasingly noise-resilient circuit ansätze grounded in the Continuous Topological Interpretation (CTI) of quantum circuits. CTI reframes quantum circuits as continuous landscapes embedded in an extended parameter space, where entanglement acts as manifold enlargement and noise as geometric deformation. Building on this foundation, we develop TOPAZ, a topology optimization framework that pre-tunes ansatz parameters before hardware execution. TOPAZ combines a tensor-dot noise engine ($O(2^N)$ per term, 3,838× speedup over dense Kraus maps), a dual-MMA optimiser that decouples intensity and timing subspaces, and an explicit two-stage sparsification protocol that actively removes entangling gates (34% depth reduction, $\Delta\text{DM} = -0.0065$). Validation spans three scales: 12-qubit LiH (beating ZNE by 1.87×), 8-qubit calibrated-device benchmarks (LPTE density-matrix fidelity 0.9085 with 21 two-qubit gates), and live IBM Heron r2 hardware (inversion test gain of 10.71× at 18σ). Across every backend and noise model, the optimiser converges to a hardware-transferable fidelity floor — the structural well predicted by CTI — confirmed by a $6 \times 6 \times 10$ noise-model transfer matrix with transfer efficiency 0.9943.

Keywords: Continuous Topological Interpretation, Parametric Trotter Expression, Quantum Topology Optimization, Noise-Resilient Quantum Circuits, Structural Well, Method of Moving Asymptotes.

1 Introduction

Variational Quantum Algorithms (VQAs) [6, 12] are the leading approach for near-term quantum computing, but they face a fundamental tension: shallow circuits limit expressivity, yet coherence times demand shallow circuits. Existing quantum error mitigation (QEM) techniques — Zero-Noise Extrapolation (ZNE) [7], Clifford Data Regression (CDR) [8], Virtual Distillation (VD) [9] — all operate on circuit outputs after execution: they post-process, fit error models, or extrapolate from amplified-noise runs [16]. None modify the circuit structure itself.

This paper introduces TOPAZ, a framework that acts *before* the circuit runs. Given a parametric Trotter ansatz, TOPAZ tunes the continuous parameters — intensity weights ρ and timing angles τ — so that the noisy output is intrinsically closer to the ideal one before any post-execution mitigation. The framework is grounded in the Continuous Topological Interpretation (CTI) [1], which reframes quantum circuits as continuous landscapes rather than discrete gate sequences.

The paper’s contributions are organized along three axes:

PTE family of operator primitives (Secs. 3–5). We derive three variants of the Parametric Trotter Expression, forming a hierarchy of increasing structure:

- **Global PTE** (Sec. 3): the original $2^N \times 2^N$ weighted-sum ansatz with polar projection — a general operator framework that closes at $N \geq 4$ due to $O(N^2/4)$ gates per polar step.
- **Product PTE** (Sec. 4): a first-order Trotter approximation that compiles to 8 two-qubit gates on 8 qubits — a proof-of-concept that TOPAZ works.

- **Local PTE (LPTE)** (Sec. 5): per-pair PTE with per-pair SVD projection, 21 two-qubit gates, and 5 effective parameters per entanglement pair — the canonical ansatz and the zenith of the PTE family.

Tensor dot engine and noise models (Sec. 6). An $O(2^N)$ engine that applies k -local Kraus operators via tensor contraction rather than dense 2^{3N} superoperator construction (3,838× speedup). We describe the six noise models against which TOPAZ is validated: the legacy triple-channel correlated model, IBM’s calibrated FakeFey and FakeMontrealV2 snapshots, and live `ibm_fey` hardware.

TOPAZ framework (Sec. 7). The framework that brings CTI to experiment: a dual-MMA optimiser with ρ/τ subspace separation, an explicit two-stage sparsification protocol that actively removes entangling gates, and algorithmic compatibility criteria. Key results include a 10.71× gain on live `ibm_fey` hardware (18σ), a $6 \times 6 \times 10$ noise-model transfer matrix with transfer efficiency 0.9943, and direct empirical observation of the structural well across every backend tested.

2 Continuous Topological Interpretation

CTI [1] provides the theoretical foundation for TOPAZ. We summarise only what is needed for the rest of the paper.

2.1 Ambient space and polar projection

Standard quantum circuits operate in the unitary group $U(d) = \{U \in \text{Mat}_d(\mathbb{C}) : U^\dagger U = I\}$, a compact Lie group

of dimension d^2 where $d = 2^N$. CTI embeds $U(d)$ as a constrained submanifold inside the full space of linear operators $\text{Mat}_d(\mathbb{C}) \cong \mathbb{C}^{d^2}$. A continuous circuit operator $A(\lambda)$ maps $\mathbb{R}^M \rightarrow \text{Mat}_d(\mathbb{C})$ without the unitarity constraint, allowing gradients to flow around barriers that would block a strictly unitary optimizer.

To execute operators on hardware, we project back to $U(d)$ via the polar decomposition. Every invertible $A \in \text{GL}(d, \mathbb{C})$ decomposes uniquely as $A = QP$ with Q unitary and $P = (A^\dagger A)^{1/2}$ positive definite [20, Sec. 8.1]:

$$\mathcal{P}_U(A) = A(A^\dagger A)^{-1/2}. \quad (1)$$

Q is the unique closest unitary to A in the Frobenius norm. This projection — **unitary regularization** — pins singular values back to 1.0, smoothing sharp wrinkles in the loss landscape and stabilizing gradient signals.

2.2 Entanglement as manifold enlargement

Without entanglement, operations are restricted to localized rotations within disjoint qubit clusters. For an N -qubit register partitioned into k clusters S_a , the set of preparable product states forms the unentangled manifold $\mathcal{M}_{\text{unt}}$ with dimension $\dim(\mathcal{M}_{\text{unt}}) = \sum_{a=1}^k d_{\text{local},a}$. Introducing entangling parameters $\lambda_{\text{ent},a}$ that bridge cluster boundaries yields the entangled manifold $\mathcal{M}_{\text{ent}}$:

$$\dim(\mathcal{M}_{\text{ent}}) = \sum_{a=1}^k (d_{\text{local},a} + d_{\text{ent},a}) > \dim(\mathcal{M}_{\text{unt}}). \quad (2)$$

Entanglement physically inflates the accessible parameter manifold, unlocking higher-dimensional navigation that the optimizer exploits to bypass local minima.

2.3 Noise as geometric deformation

Environmental noise (T_1 , T_2 , readout error) acts as a continuous vector field pulling the state landscape away from the target manifold. Instead of correcting the output, TOPAZ searches for a **structural well** — a region of the parameter manifold where the noise field and control gradients approximately cancel. By pre-tuning (ρ, τ) to sit inside this well, noise resilience is built directly into the circuit’s structural parameters.

3 Global PTE

The Parametric Trotter Expression (PTE) is a parameterized operator ansatz where the free variables correspond to gate intensities and timings rather than rotation angles. Unlike conventional VQA parameterisations that maintain strict unitarity at every step, PTE allows the operator to temporarily leave the unitary group, explore a larger manifold, and project back via polar decomposition — a feature that expands expressivity per gate [3].

3.1 Derivation from Trotter–Suzuki

For a Hamiltonian $H = \sum_{j=1}^M H_j$ decomposed into Hermitian (typically two-local Pauli) terms, the Trotter–Suzuki approxi-

mation [19, 13] splits time evolution into n slices:

$$e^{-iHt} = \lim_{n \rightarrow \infty} \left[\prod_{p=1}^n \prod_j \exp(-iH_j \delta) \right], \quad \delta = \frac{t}{n}. \quad (3)$$

At finite n the error scales as $O(t^2/n)$ [17].

We extend the Trotter decomposition by introducing two classes of design variables for each term j and layer k : intensity weights $\rho_{j,k} \in [0, 1]$ with $\sum_j \rho_{j,k} = 1$, and timing scaling factors $\tau_{j,k} \in [0, \tau_{\text{max}}]$. The layer- k PTE operator is

$$B_k(\rho, \tau) = \sum_{j=1}^M \rho_{j,k} e^{-iH_j \delta \tau_{j,k}}, \quad (4)$$

and the full PTE operator is

$$A(\rho, \tau) = \prod_{k=1}^n B_k(\rho, \tau). \quad (5)$$

Route (A) — applying weights directly to the Hamiltonian, $e^{-i \sum_j \rho_j H_j \delta}$ — maintains unitarity by construction but limits expressivity to weighted Hamiltonians. Route (B), the PTE, creates weighted sums of exponentials. The distinction is fundamental: the sum of unitaries in Eq. (4) is generically non-unitary, which is the source of PTE’s expressive power.

3.2 Non-unitarity and polar projection

A weighted sum of unitary operators is generally non-unitary. Consider $B = \sum_j \alpha_j U_j$:

$$B^\dagger B = \sum_{j,k} \alpha_j^* \alpha_k U_j^\dagger U_k. \quad (6)$$

For $B^\dagger B = I$ we require the cross terms to cancel — a condition satisfied only when all but one $\alpha_j = 0$ or under special symmetric constructions. This non-unitarity is not a flaw: the mathematical freedom to leave the unitary manifold enables the exploration of operator spaces unreachable by traditional variational circuits.

We restore physical realizability via layer-wise polar projection. For each layer k :

$$U_k = B_k (B_k^\dagger B_k)^{-1/2}, \quad U_{\text{safe}}(\rho, \tau) = \prod_{k=1}^n U_k. \quad (7)$$

We compute U_k via SVD with a Newton–Schulz refinement step and a final determinant correction $U_{\text{safe}} \rightarrow U_{\text{safe}} \cdot (\det U_{\text{safe}})^{-1/d}$ to remove the global phase.

3.3 Scope and limitations

The Global PTE is the original formulation [3], requiring $O(N^2/4)$ two-qubit gates per polar step and thus intractable for $N \geq 4$. Its inversion test on 8 qubits registers 0.004, equal to the noise floor $1/256 \approx 0.0039$. The structural lower bound on quantum-state decomposition permanently closes this line for multi-qubit systems. The Product PTE and LPTE variants that follow resolve this by introducing locality.

3.4 Objective functions

The optimizer minimises the noisy fidelity loss:

$$L_{\text{op}}(\rho, \tau) = 1 - \frac{1}{d^2} |\text{Tr}(U_{\text{target}}^\dagger U_{\text{safe}}(\rho, \tau))|^2, \quad (8)$$

or the state-fidelity loss for a target state $|\psi_{\text{target}}\rangle$:

$$L_{\text{st}}(\rho, \tau) = 1 - |\langle \psi_{\text{target}} | U_{\text{safe}}(\rho, \tau) | \psi_0 \rangle|^2. \quad (9)$$

A regularisation term keeps the pre-projection operator close to unitarity:

$$L(\rho, \tau) = L_{\text{prim}}(\rho, \tau) + \lambda \sum_{k=1}^L \|B_k^\dagger B_k - I\|_F^2, \quad \lambda = 10^{-3}. \quad (10)$$

4 Product PTE

The Global PTE’s $2^N \times 2^N$ construction cannot scale. The Product PTE replaces the dense matrix product with a first-order Trotter product that compiles term-by-term to hardware-native gates. For an N -qubit system with M Hamiltonian terms:

$$U_{\text{circ}}(\rho, \tau) = \prod_{j=1}^M e^{-i \rho_j H_j \delta \tau_j}, \quad (11)$$

where each factor is a k -local unitary ($k = 2$ for nearest-neighbour Pauli terms). The polar projection is applied per term via SVD, but the product structure eliminates the need for layer-wise $2^N \times 2^N$ reconstruction.

On 8 qubits with $M = 21$ nearest-neighbour Pauli terms (seven $\{XX, YY, ZZ\}$ triples), the Product PTE compiles to 8 two-qubit gates and achieves DM fidelity 0.6329 under FakeFez noise — a 9.35 $\times$ gain over random baseline (0.0677). The circuit has 42 parameters and no pruning mechanism (self-pruning is absent in the product form because the Trotter factors are individually unitary).

The Product PTE serves as a proof-of-concept: it establishes that the TOPAZ optimisation pipeline works and that the structural well is reachable even with a simple ansatz. However, its lack of self-pruning and limited fidelity make it a stepping stone to LPTE, which dominates it on every metric (Sec. 5).

5 Local PTE (LPTE)

The Local PTE (LPTE) is the zenith of the PTE family. It combines the expressivity of the Global PTE with the locality of the Product PTE by applying a per-pair PTE sum with per-pair SVD projection — retaining the weighted-sum structure that enables self-pruning while keeping the circuit depth tractable.

5.1 Construction

For each nearest-neighbour pair (q_i, q_{i+1}) in an N -qubit linear chain, we define a local PTE operator:

$$B_{(i,i+1)}(\rho, \tau) = \sum_{P \in \{XX, YY, ZZ\}} r_{i,P} e^{-i P \delta \tau_{i,P}}, \quad (12)$$

Table 1. Full ablation study on the 8-qubit FakeFez benchmark using the LPTE ansatz (21 two-qubit gates in full configuration). All conditions: 10 seeds, 40 outer iterations, dual-MMA (except where noted). Significance vs. full TOPAZ: Bonferroni-corrected $\alpha = 0.05/36 \approx 0.0014$.

Condition	DM Fid.	2Q Gates	Active	Cohen’s d
Exact preparation (oracle)	0.924 ± 0.003	21.0	21.0	26.43***
Full TOPAZ (baseline)	0.737 ± 0.066	12.6	9.2	—
COBYLA (LPTE)	0.729 ± 0.069	15.6	13.6	0.12 (ns)
Single MMA	0.623 ± 0.166	19.9	19.9	0.80 (ns)
Adam (LPTE)	0.514 ± 0.209	16.5	14.8	1.02 (ns)
No polar decomposition	0.334 ± 0.170	16.7	15.8	2.14***
HEA (3-layer)	0.329 ± 0.110	28.0	37.2	3.97***
Product-only PTE	0.279 ± 0.175	16.7	16.0	3.09***
Random PTE	0.056 ± 0.031	21.0	21.0	8.69***

*** $p < 0.001$ (Bonferroni-passing at $\alpha_{\text{Bonf}} = 0.0014$). Cohen’s d vs. full TOPAZ.

with intensities $r_{i,P} \in [0, 1]$ and timings $\tau_{i,P} \in [0, \tau_{\text{max}}]$, normalised so $\sum_P r_{i,P} = 1$. Each $B_{(i,i+1)}$ is projected via SVD to its nearest unitary $U_{(i,i+1)}$. The full LPTE operator is the product of these per-pair unitaries:

$$U_{\text{LPTE}}(\rho, \tau) = \prod_{i=0}^{N-2} U_{(i,i+1)}. \quad (13)$$

On 8 qubits with 7 nearest-neighbour pairs and 3 Pauli terms per pair, LPTE uses 21 two-qubit gates and 42 parameters — the same parameter count as Product PTE but 2.6 $\times$ the gate count. The key difference is structural: the per-pair SVD projection in LPTE allows the optimizer to discover which pairs contribute to noise resilience and assign them nonzero intensity, while Product PTE’s purely unitary Trotter factors lack this selection mechanism.

5.2 Self-pruning

The central empirical discovery is that LPTE **self-prunes** under noise. During optimization, the MMA solver drives the ρ gradients of low-value pairs below the clipping floor $\max(\rho, 10^{-6})$, where the gradient vanishes and the parameters remain frozen. Across 10 seeds under FakeFez, typically 8–24 of 42 (ρ, τ) dimensions survive, corresponding to ~ 9 of 21 entangling pairs actively contributing. Table 1 shows the 9-condition ablation confirming that self-pruning requires the full TOPAZ stack (PTE ansatz + polar projection + dual-MMA).

The key findings: Full TOPAZ (DM 0.737 ± 0.066) significantly beats product-only PTE (0.279 ± 0.175 , $p < 0.001$, $d = 3.09$), HEA (0.329 ± 0.110 , $p < 0.001$, $d = 3.97$), and the no-polar-decomposition variant (0.334 ± 0.170 , $p < 0.001$, $d = 2.14$). The polar projection is critical — removing it collapses fidelity by over half.

5.3 Parameter efficiency

Table 2 compares PTE-family parameter efficiency against HEA on 8 qubits under FakeFez. LPTE achieves 0.9085 DM fidelity with 21 two-qubit gates and 42 parameters, while HEA uses ~ 96 parameters and 28 gates for 0.1517 — a 6 $\times$ gap in fidelity per parameter.

The inductive bias is the reason: PTE’s parameterisation mirrors the Hamiltonian structure (each (ρ_j, τ_j) ties directly to a specific Hamiltonian term), so the ansatz starts in the

Table 2. Parameter efficiency: PTE-family vs. HEA on 8 qubits under FakeFez.

Ansatz	Parameters	2Q Gates	DM Fid.
PTE (product)	42	8	0.6329
LPTE	42	21	0.9085
HEA (3-layer)	~ 96	28	0.1517

Table 3. Per-evaluation runtime at $N = 10$ for one layered triple-channel Kraus map.

Engine	Time (s)	Complexity per term
Qiskit dense Kraus map	180.400	$O(2^{3N})$
Tensor-dot (this work)	0.047	$O(2^N)$
Speedup	~3838×	—

right functional ballpark. An HEA of comparable gate count must burn parameters learning which qubits should talk to each other.

6 Tensor Dot Engine and Noise Models

6.1 Tensor-dot noise engine

Applying a sequence of k -local Kraus operators to a dense density-matrix tensor normally requires constructing the full $2^{2N} \times 2^{2N}$ superoperator — complexity $O(2^{3N})$. The tensor-dot engine reframes ρ as a tensor with $2N$ indices and performs local contractions:

$$\rho'_{i'_S, i_{S^c}; j'_S, j_{S^c}} = \sum_a K_{i'_S, i_S}^{(a)} \rho_{i_S, i_{S^c}; j_S, j_{S^c}} K_{j_S, j'_S}^{(a)\dagger} \quad (14)$$

where S is the active qubit set and S^c are spectator qubits. This reduces complexity to $O(2^N)$ per term.

The engine plays different roles on different experiment paths. On the legacy triple-channel path (LiH, algorithm tests), it handles both the ideal-state computation and noisy density-matrix evolution. On the FakeFez simulator path, its role is narrowed to the noiseless target state; noisy evaluations are delegated to Qiskit’s AerSimulator. On live `ibm_fez` hardware, the engine has no role — the physical backend executes transpiled gates directly.

6.2 Noise models

TOPAZ is validated against six distinct noise profiles, spanning the gap from idealized toy models to live hardware:

Triple-channel correlated (legacy). Three sequential channels: depolarising at $p = 0.005$, amplitude damping at $\gamma = 0.008$, phase damping at $\lambda = 0.012$, each with single-qubit and multi-qubit components. Multi-qubit correlations decay exponentially with qubit distance, $p_{\text{corr}}(d) = \alpha p e^{-d/\xi}$ with $\xi = 2.0$. The Kraus operators are standard [18]. This model captures the qualitative shape of NISQ noise without being tied to any specific device.

FakeFez. IBM’s calibrated noise snapshot of the live `ibm_fez` processor (Heron r2, 156 qubits), carrying per-gate error rates, qubit-specific T_1 and T_2 times, readout errors, and

the device coupling map [10]. All headline experiments use FakeFez as the noise model during optimisation.

FakeMontrealV2. A 27-qubit device-calibrated noise model used for the earlier port verification. A 3-iteration smoke test moved from 7.4% to 36.0% DM fidelity, confirming the optimiser finds productive descent directions quickly under full device noise.

Live `ibm_fez` hardware. The optimised (ρ, τ) from FakeFez are submitted to the live 156-qubit Heron r2 [11] processor for the inversion test. No re-optimisation is performed on hardware; the parameters transfer directly.

The six-model transfer matrix (Sec. 8.7) demonstrates that the structural well is noise-model agnostic: parameters optimised under any model perform at 99.4% of their on-model fidelity when evaluated under a different model.

7 TOPAZ Framework

7.1 Dual-MMA optimiser

The MMA algorithm [4, 5] constructs a strictly convex separable approximation of the objective at each iteration and solves it analytically subject to asymptotic bounds. The dual-MMA optimiser has two meanings that travel together:

Meaning 1: subspace separation. The intensity vector ρ and timing vector τ are heterogeneous parameter classes whose finite-difference step sizes differ by roughly four orders of magnitude ($\epsilon_\rho = 10^{-4}$ vs. $\epsilon_\tau = \pi/4$). A single MMA instance with a shared move-limit budget systematically misallocates the parameter-update budget. Dual-MMA runs two independent MMA solvers — one for ρ , one for τ — with their own asymptote, move-limit, and momentum state. The momentum-smoothed gradient:

$$\nabla_{\text{smooth}} = \alpha \nabla_{\text{cur}} + (1 - \alpha) \nabla_{\text{prev}}, \quad \alpha = 0.4, \quad (15)$$

stabilises convergence. Move limits adapt online based on parameter change magnitude.

Meaning 2: stacked modules across parameter classes. When a target algorithm carries an additional class of parameters (e.g., VQE rotation angles), TOPAZ replaces the classical optimiser with a second MMA module wrapped around its own small PTE ansatz. The two modules alternate, each treating the other’s parameters as fixed. This paper exercises two modules (structural + algorithmic); a third class slots in the same way.

7.2 Topology optimization via sparsification

The LPTE ansatz defines an entangling graph $G_{\text{LPTE}}(\rho, \tau) = (V, E)$ where each pair p contributes a 4-qubit unitary with normalised intensity $\kappa(p) = \sum_P r_{p,P}$. Passive self-pruning discovers a sparse subspace within the continuous parameter domain. We formalize an **active** two-stage protocol that removes entangling gates from the circuit:

Stage 1 — Continuous relaxation. Solve $\min_{\rho, \tau} \mathcal{L}(\rho, \tau)$ on E_{max} with $\rho \geq 0$ and no artificial floor. The optimizer naturally drives low-value pairs toward zero.

Table 4. Multi-seed topology optimization benchmark ($\kappa = 0.05$, 5 seeds). F = full (Stage 1), S = sparse (Stage 3).

Seed	DM fidelity		2Q gates		Depth		Pruned
	F	S	F	S	F	S	
0	0.8487	0.8291	20	18	79	52	1
1	0.7456	0.7324	17	15	63	40	1
2	0.8910	0.8910	21	21	85	85	0
3	0.5220	0.5301	15	9	58	24	3
4	0.6802	0.6726	20	17	77	39	1

Stage 2 — Hard pruning. Remove pairs with $\kappa(p) < \tau_\kappa$ (pre-registered threshold $\tau_\kappa = 0.05$):

$$E_{\text{pruned}} = \{(q_i, q_{i+1}) \mid \kappa(p) \geq \tau_\kappa\}. \quad (16)$$

Stage 3 — Sparse re-optimization. Solve $\min_{\rho', \tau'} \mathcal{L}(\rho', \tau' \mid E_{\text{pruned}})$ from fresh random restarts on the reduced domain.

Across 5 seeds (Table 4), the pruned circuit achieves 14% gate reduction (18.6 $\rightarrow$ 16.0), 34% depth reduction (72.4 $\rightarrow$ 48.0), and mean DM fidelity loss of only -0.0065 (0.7375 $\rightarrow$ 0.7310). Seed 3 removes three of seven pairs, cutting gates by 40% and depth by 59% while actually *improving* DM fidelity by $+0.0081$. The 34% depth reduction compounds on hardware, where fewer sequential gates mean shorter coherence exposure.

7.3 Algorithmic compatibility

Whether continuous parameter optimisation works on a given algorithm depends on four topological properties of its fidelity landscape: connectedness, compactness, continuity, and convexity. Algorithms satisfying all four — VQE, VQA, QAOA, parametrised quantum simulation — are natural TOPAZ targets. Grover’s algorithm as a whole violates continuity and convexity; the Grover oracle layer is compatible via a topological lift (3.09 $\times$ gain), establishing the principle that non-differentiable algorithms can be protected by applying TOPAZ to their structural primitives.

8 Results

8.1 8-qubit trajectories under triple-channel noise

Configuration: 8 qubits, 9 nearest-neighbour Pauli terms, 18 parameters, 40-iteration cap. Pre-optimisation noisy fidelity ranges 0.07–0.70; post-MMA fidelity rises to 0.30–0.88, an absolute improvement of 18–23 pp. The ~ 47 pp spread in final fidelity reflects a multi-modal landscape (the dual-MMA navigates several local optima at different floor depths).

8.2 H₂ VQE on FakeFez

At 2 qubits (STO-3G basis, UCCSD ansatz), TOPAZ reduces the noisy UCCSD baseline of 70.6 ± 2.3 mHa to 12.8 mHa (5.5 $\times$ improvement). The result is consistent with the structural-well picture: the noise-resilient basin appears at 2 qubits as it does at 8 and 12.

[Figure: Two representative 8-qubit optimisation trajectories]

Figure 1. Two representative 8-qubit trajectories from independent random seeds under triple-channel correlated noise. Run 1: 0.172 $\rightarrow$ 0.371; Run 2: 0.233 $\rightarrow$ 0.838.

Table 5. H₂ VQE benchmark under FakeFez.

Metric	Value
Noisy UCCSD baseline	70.6 ± 2.3 mHa
TOPAZ	12.8 mHa
Chemical precision	1.6 mHa
Improvement vs. baseline	5.5 $\times$

Table 6. H₂ VQE SOTA benchmark under FakeFez.

Ansatz	ΔE_{FCI} (mHa) noisy	ΔE_{FCI} ZNE+CDR (mHa)	Purity	Params	Conv.
LPTE (dual-layer)	271 ± 271	247 ± 277	0.961 ± 0.020	36	60%
HEA (depth 3)	48.7 ± 5.6	0.0*	0.935 ± 0.005	28	100%
HVA ($d = 2$)	1582 ± 13	2177 ± 19	0.451 ± 0.005	30	100%
HVA ($d = 4$)	1583 ± 13	2171 ± 10	0.452 ± 0.008	60	100%

*Clamped to FCI; see text for discussion.

8.3 4-qubit H₂ SOTA benchmark

This is the **Blind TOPAZ** benchmark (so called because TOPAZ is used purely as a variational ansatz with no knowledge of the QEM post-processing — the optimiser sees only noisy energies, never clean reference values): the LPTE ansatz, tested as a variational circuit for energy-based VQE under FakeFez (FCI reference -1.13730604 Ha). Three ansätze are compared: dual-layer LPTE (36 parameters, optimised with MMA for 60 iterations), HEA depth 3 (28 parameters, COBYLA 500 iterations), and HVA $d = 2$ and $d = 4$ (30 and 60 parameters). All run for 10 independent random seeds with ZNE, CDR, and VD post-processing.

LPTE plus ZNE+CDR reaches $\Delta E_{\text{FCI}} = +1.78$ mHa on the best seed (seed 9) — effectively chemical accuracy. Across 10 seeds, the good plateau (6 of 10) achieves $\Delta E_{\text{FCI}} = 21.7 \pm 21.0$ mHa under ZNE+CDR; the remaining 4 seeds converge to a local minimum at $\Delta E_{\text{FCI}} \approx 585$ mHa. Critically, the purity is indistinguishable between plateaus (0.961 ± 0.019 vs. 0.962 ± 0.019), confirming the failure mode is classical optimisation, not quantum noise.

The HEA ZNE+CDR entry in Table 6 is marked with an asterisk because the second-order polynomial ZNE extrapolation produced a negative ΔE_{FCI} — a known artifact when the noise curve deviates from the assumed exponential model. We clamp all ZNE and CDR values at FCI before computing statistics. The clamping rate itself is a diagnostic: HEA required clamping on 8 of 10 seeds, while TOPAZ/LPTE required clamping on *none*. That TOPAZ’s extrapolations never violated the physical bound implies its noise-energy relation-

Table 7. LiH ground-state benchmark, 10 seeds, triple-channel correlated noise.

System scale	FCI target	Noisy baseline error	TOPAZ error (10 seeds)
4-qubit (active space)	-7.8823 Ha	~275.4 mHa	25.63 $\pm$ 2.64 mHa
10-qubit	-7.8823 Ha	>1000 mHa	30.78 $\pm$ 0.97 mHa
12-qubit (full JW)	-7.8823 Ha	>1000 mHa	30.78 $\pm$ 0.97 mHa

Table 8. Cross-algorithm benchmark under triple-channel noise.

Algorithm	Qubits	Baseline	TOPAZ	Gain
Bernstein–Vazirani	4	40.96%	75.88%	1.85×
Grover (oracle layer)	4	22.00%	68.00%	3.09×
Chaos Trotter	8	12.40%	84.39%	6.81×

Table 9. Comparison on FakeFез (quietest path, 2000 shots). LPTE beats exact preparation in both DM and inversion test fidelity while using half the gates.

Method	DM Fid.	Inv. test fid.	CDR Inv.	VD Inv.	2Q Gates	Depth
Exact prep (raw)	0.8501	0.7150	0.2313	0.9930	42	195
Exact + ZNE	0.9813*	0.7308	—	—	42	195
HEA (Ry+CX, 4 layers)	0.1517	0.1305	—	—	28	57
TOPAZ product	0.6329	0.5370	—	—	8	22
TOPAZ LPTE	0.9085	0.7715	0.9349	0.9964	21	97
TOPAZ LPTE + ZNE	0.9743*	0.7917	—	—	21	97

*ZNE DM is extrapolated to $\lambda = 0$ via exponential fit, not measured directly.

ship is closer to the assumed exponential form — a direct consequence of the structural well pre-tuning the circuit into a region where the noise model is better-behaved.

8.4 12-qubit LiH ground state

The full Jordan–Wigner mapping at 12 qubits, bond distance 1.6 Å, under triple-channel correlated noise. The dual-MMA architecture exercises both meanings: the structural module tunes (ρ, τ) , and the algorithmic module wraps a second MMA solver around the VQE rotation angles.

The 12-qubit mean error of 30.78 mHa is reached in roughly 22 iterations with 100% convergence across seeds. The error margin barely grows with system size, consistent with the structural-well picture.

8.5 Cross-algorithm: BV and Grover

The Grover result is the most revealing. Earlier work reported Grover as incompatible with TOPAZ because amplitude amplification fails the continuity and convexity criteria. Applying TOPAZ only to the diffusion-oracle implementation — treating the oracle as a PTE landscape while leaving the Boolean search logic untouched — recovers a 3.09× resilience advantage. The principle generalises: non-differentiable algorithms can be protected by acting on their structural primitives.

8.6 Validation on FakeFез

All experiments in this subsection use the 8-qubit Trotter ansatz with $M = 21$ nearest-neighbour Pauli terms on the quietest ibm_fez path ([127, 137, 147, 146, 145, 144, 143, 136], path mean two-qubit gate error 0.31%). Optimisation runs against FakeFез; the same (ρ, τ) are then submitted to live hardware.

[Figure: Canonical FakeFез optimisation trajectory]

Figure 2. Canonical FakeFез optimisation trajectory: steep early descent in 5–10 iterations, then convergence to a stable floor at $F \approx 0.6329$ (Product PTE).

Table 10. Per-model DM fidelity under six noise models.

Model	DM μ	DM σ	DFE μ	Active μ
Nominal (triple-channel)	0.3985	0.1927	0.3379	20.3
Low noise (50% amplitude)	0.3710	0.1837	0.3397	20.2
High depolarisation (2×)	0.3605	0.1869	0.3029	20.3
Amplitude damping (T_1)	0.2696	0.1342	0.1962	19.7
Phase damping (T_2)	0.4150	0.1835	0.3206	19.5
High readout (5%/8%)	0.3985	0.1927	0.2699	20.3

Transfer efficiency (off-diagonal / diagonal mean): 0.9943. All pairwise $p > 0.07$.

Noiseless fidelity decomposition. LPTE achieves $F_{\text{ideal}} = 0.9829$ (approximation loss 1.71%); under noise it records $F_{\text{noisy}} = 0.9085$ (noise loss 7.41%). Exact preparation loses 14.99% under identical noise (0.8501 from 1.0). LPTE is roughly 2× more resilient to noise than the exact unitary, and the approximation penalty is only 1.71% of the gap to perfect fidelity. The advantage is dual-mechanism: the structural well absorbs noise more effectively, and the polar projection stabilises optimisation.

Multi-seed stability. Across $n = 10$ independent runs under FakeFез, the optimiser converges to DM 0.739 ± 0.066 , inversion test 0.552 ± 0.044 , and 9.0 ± 1.3 active terms out of 21. Every seed prunes to 7–12 active terms — never all 21 — confirming self-pruning is systematic.

8.7 Noise-model transfer matrix

Parameters optimised under one noise model were cross-evaluated on five others, producing a $6 \times 6 \times 10$ transfer matrix (360 evaluations). The six models are: nominal (triple-channel), low-noise (50% amplitude), high-depolarisation (2×), amplitude-damping, phase-damping, and high-readout (5%/8%).

The critical result is the transfer efficiency: parameters optimised under one noise model perform at 99.4% of their on-model fidelity when evaluated under a different model (Fig. 3). The structural well is not an overfit to a specific noise profile but a generic geometric property of the LPTE ansatz under structured noise. This has a practical corollary: parameter sets optimised in simulation transfer to hardware without re-optimisation.

[Figure: 6x6 noise-model transfer matrix]

Figure 3. 6x6 noise-model transfer matrix. Transfer efficiency: 0.9943.

Table 11. Real-hardware validation. Errors on inversion test values are 1σ shot-noise estimates from 2000 shots; noise floor is $1/2^8 \approx 0.0039$.

Method	Baseline	TOPAZ	Gain
FakeFez DM (exact)	0.0677	0.6416	9.48×
FakeFez + ZNE DM	0.0747	0.6607	8.85×
ibm_fez Inv. test (live)	0.0155 ± 0.0028	0.1660 ± 0.0083	10.71×
ibm_fez Inv. (path-matched)	0.2960 ± 0.0102 (exact)	0.3610 ± 0.0107 (LPTE)	4.4 σ

Table 12. 8-qubit mitigation stacking on FakeFez (LPTE baseline).

Configuration	DM Fid.	Inv. fid.
Baseline (no mitigation)	0.0677	0.0525
TOPAZ LPTE alone	0.9079	0.7715
LPTE + ZNE	0.9746	0.7692
LPTE + CDR	—	0.9349
LPTE + VD	—	0.9964

8.8 Real hardware: ibm_fez

The optimised (ρ, τ) from the FakeFez benchmark (Sec. 8) were submitted to live ibm_fez (Heron r2, 156 qubits) for the inversion test with 2000 shots.

Total QPU time was ~ 2 minutes 15 seconds. The 10.71× gain is preserved at 18σ significance. After correcting a transpiler-layout bug that pinned circuit qubits to physical qubits 0–7 instead of the quietest path, the path-matched follow-up raised absolute fidelities to 0.2960 ± 0.0102 (exact) and 0.3610 ± 0.0107 (LPTE), a 4.4σ separation.

Why the inversion test is lower than DM. The inversion test runs $2\times$ the gate count; the extra noise suppresses it by $\sim 74\%$ relative to DM. The gap between simulator and hardware (0.64 vs. 0.16) decomposes into path mismatch ($\sim 40\%$), unmodelled real-hardware noise ($\sim 35\%$), and shot noise ($\sim$ small, $\sigma \approx \pm 0.008$). The 10.71× gain ratio is the honest metric because both baseline and TOPAZ suffer identical real-hardware overhead proportionally.

8.9 Stacking with ZNE, CDR, VD

TOPAZ composes multiplicatively with post-processing techniques because they act on different parts of the pipeline: TOPAZ pre-tunes parameter space; ZNE, CDR, and VD extrapolate or correct observables.

The relevant comparison is not “TOPAZ vs. ZNE” but “what does the stack look like”. ZNE alone at the baseline

Table 13. SOTA comparison on the 12-qubit LiH benchmark.

Technique	Mitigated error [mHa]	Recovery R	E_s/E_T
ZNE (Richardson)	50.79	81.6%	1.87×
CDR	69.09	74.9%	2.54×
Virtual Distillation (2-copy)	78.61	71.5%	2.89×
TOPAZ (this work)	27.20	90.1%	1.00×

[Figure: The structural well on FakeFez, 2D (ρ, τ) slice]

Figure 4. The structural well on FakeFez, projected onto a 2D (ρ, τ) slice. Darker regions indicate lower loss $(1 - F)$.

gives marginal improvement ($0.0677 \rightarrow 0.0747$). On top of TOPAZ, ZNE contributes a further 0.0667 DM gain. TOPAZ moves the operating point onto a part of the noise curve where ZNE’s extrapolation is more reliable.

8.10 Head-to-head against ZNE, CDR, VD

On the 12-qubit LiH benchmark under triple-channel noise, TOPAZ is compared against three established QEM techniques under matched conditions where possible.

TOPAZ beats ZNE by 1.87×, CDR by 2.54×, and VD by 2.89×. The two techniques are not in conflict: they stack (Table 12), but if you can only run one, TOPAZ delivers more.

9 The Structural Well

Across every seed, session, and backend — triple-channel noise, FakeFez, FakeMontrealV2, live ibm_fez — the optimisation trajectory shows the same canonical shape: a rapid fidelity descent in the first 5–10 iterations, then convergence to a stable floor. The companion CTI paper [1] predicted this: noise-deformed parameter landscapes contain **structural wells** — basins where the noisy output is intrinsically robust.

Table 14. Hessian summary of the structural well on FakeFez.

Metric	Value
Simulations per landscape	3,529
Time per landscape	~ 4 min
κ_ρ	~ 1
κ_τ	$\sim 9 \times 10^3$
Combined κ	$\sim 2 \times 10^5$
Active dimensions	8–24

9.1 Properties

The well has four observable properties:

- **Attractive.** Reached from arbitrary random initialisations.
- **Stable.** The floor value reproduces across seeds and sessions.
- **Hardware-transferable.** The same basin appears under live `ibm_fez` noise as under FakeFez.
- **Entanglement-dependent.** A product-state circuit with the same gate count cannot reach a comparable floor.

Multi-seed convergence confirms the attractor: 10 random initialisations produce mean DM 0.739 with $\sigma = 0.066$, all within 2σ of the floor.

9.2 Sparse subspace mechanism

The well is **not** a 3D attractor. It lives in whatever sparse subspace the optimisation discovers — typically 8–24 effective dimensions from the 42-dimensional (ρ, τ) space. The sparsity mechanism is gradient clipping: $\nabla_{\rho, \tau} \max(\rho, 10^{-6})$ vanishes below threshold, so pruned parameters stay pruned.

Crucially, the survivors are *not* the target state’s largest coefficients. The top six target-Hamiltonian terms all get pruned; the optimiser selects for noise-robustness, not target-matching. This sparsity was not predicted by CTI-E, which anticipated a geometric basin but not the gradient-clipping selection mechanism.

9.3 Canyon-river curvature

The Hessian at the well reveals a canyon-river landscape:

- $\kappa_\rho \approx 1$ (all ρ equally stiff at ~ 870)
- $\kappa_\tau \approx 9 \times 10^3$ (timing varies by 8,000 $\times$ across survivors)
- Combined $\kappa \approx 2 \times 10^5$

The intensity subspace is uniformly steep; the timing subspace has a few extremely stiff directions amidst many soft ones. The polar projection may assist optimisation by filtering non-physical directions that contribute mostly variance in a standard rotation-angle parameterisation. The Hessian eigenvalue clustering (two sharp bands rather than a continuous spectrum) is *consistent with* dampened barren-plateau behaviour [14], but a direct diagnostic (e.g. eigenvalue broadening vs. qubit count) has not been performed. The claim here is limited to consistency, not a rigorous proof of barren-plateau avoidance [15].

9.4 Transferability

The noise-model transfer matrix (Sec. 8.7) with efficiency 0.9943 confirms the well is not an overfit to any specific noise profile. Parameters optimised under one model work essentially unchanged under any other. This is the strongest evidence that the structural well is a generic geometric property of the LPTE ansatz under structured noise — and the reason TOPAZ parameters transfer from simulation to hardware without re-optimisation.

10 Discussion and Conclusion

Summary of contributions. This paper presents a complete hierarchy: CTI provides the theoretical foundation (noise as deformation, entanglement as manifold enlargement); the PTE family (Global $\rightarrow$ Product $\rightarrow$ LPTE) provides the operator primitives, with LPTE as the zenith combining expressivity, locality, and self-pruning; the tensor-dot engine makes multi-seed optimisation tractable; and TOPAZ brings the framework to experiment with dual-MMA, sparsification, and hardware execution. The structural well — confirmed across six noise models and live hardware — is the experimental signature of CTI’s central prediction.

The pre-execution paradigm. TOPAZ extends the earlier framework from simulator-only validation to calibrated device-noise models and live IBM hardware. On 12-qubit LiH, it achieves 30.78 ± 0.97 mHa, beating ZNE by 1.87 $\times$. On FakeFez, LPTE DM fidelity is 0.9085 with 21 two-qubit gates — surpassing exact preparation (0.8501) while using half the gates. On live `ibm_fez`, the inversion test rises 10.71 $\times$ at 18σ . The optimiser converges to the same hardware-transferable floor from any random start, confirming the well is a feature of the ansatz-noise landscape, not an artefact of the optimiser.

This is not just a faster error mitigation. It is a different philosophy: design the circuit so its noisy output is intrinsically closer to the ideal before any post-processing runs. The structural well is generic — reached from any random start, stable across seeds, transferable across noise models. Noise-aware ansatz tuning becomes a routine pre-execution step: run TOPAZ once per hardware configuration, save the optimal (ρ, τ) vector, and reuse it.

TOPAZ		framework	map.
Component	Function	Sec.	Headline result
CTI	Theoretical foundation	II	Structural well prediction
Global PTE	Full $2^N \times 2^N$ operator, polar projection	III	Foundation; closed at $N \geq 4$
Product PTE	First-order Trotter product (PoC)	IV	0.6329 DM, 8 gates, 42 params
LPTE	Per-pair PTE, SVD, self-pruning	V	0.9085 DM, 56% pruned
Tensor-dot engine	$O(2^N)$ Kraus-map contraction	VI	3.838 $\times$ speedup
Dual-MMA	ρ/τ subspace decoupling	VII	~ 10 iterations to convergence
Sparsification	Active gate pruning + re-optimisation	VII	34% depth red., $\Delta DM = -0.0065$
FakeFez validation	Calibrated-noise benchmark	VIII	DM 0.737 ± 0.066 ($n = 10$)
Noise transfer	6 $\times$ 6 cross-evaluation	VIII	Transfer efficiency 0.9943
IBM Fez hardware	Live inversion test on Heron r2	VIII	10.71 $\times$ gain (18σ)
Structural well	Noise-robust attractor	IX	Sparse (8–24 dims), generic

Table 15. TOPAZ framework summary.

Near-term outlook. Immediate next steps include benchmarking higher-order Suzuki splittings, stacking with dynamical decoupling and PEC on live hardware, scaling beyond 20 qubits with matrix-product-state extensions, and full in-house CDR/VD integration matched to TOPAZ’s physical baseline.

Data and code availability

Code is available at github.com/KrishanuKarun/TOPAZ-Noise-models-and-applications, with the primary hardware notebook at python/ibm_fez_lpte.ipynb.

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